

Discrete Hexagonal Control Volumes for Non-Equilibrium Planetary Thermodynamics: Formalizing the H3 Spatial State Tensor

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<https://github.com/pascalranoroarijaona/WebOfLife>

September 14, 2026

Abstract

Planetary-scale ecological and biophysical simulations require the strict mathematical unification of discrete geodesic spatial geometries with non-equilibrium thermodynamic state laws. In this work, corresponding to Sprint 042 of the open-source *Web of Life* planetary simulation engine, we formalize discrete hexagonal control volumes over the Uber H3 discrete global grid system (DGGs). We introduce the `H3CellThermodynamicState` interface and its immutable record realization `H3CellThermodynamicRecord` in `src/spatial/h3_state_tensor.ts`. Every hexagonal cell column is formalized as an open thermodynamic control volume with explicit scalar representations of physical state properties, including internal thermal energy (U_c), absolute temperature (T_c), dynamic composite heat capacity ($C_{v,c}$), blackbody Stefan-Boltzmann dissipation ($\Phi_{\text{lw}}^{\text{out}}$), sensible and latent turbulent fluxes (Φ_{sens} , Φ_{lat}), multiphase water stocks, conserved elemental nutrients (C, N, P), and internal entropy production rate (σ_c). We demonstrate that encapsulating physical transformations within pure monadic kernels (`SpatialMonad`) guarantees the First Law conservation of mass and energy as well as Second Law non-negative local entropy production ($\sigma_c \geq 0$) under radiative solar degradation and inter-cell advection.

1 Introduction and Theoretical Motivation

Non-equilibrium thermodynamics provides the ultimate physical framework for systems ecology and planetary homeostasis [1, 2, 3]. The terrestrial biosphere is driven by a continuous exergy gradient: high-temperature shortwave solar irradiance ($T_{\text{sun}} \approx 5778 \text{ K}$) is absorbed at the planetary surface and atmospheric column, converted into chemical work, mechanical motion, and latent heat, and subsequently re-radiated as low-temperature infrared photons ($T_{\text{space}} \approx 2.725 \text{ K}$) into deep space [4].

Despite the universality of these principles, computational Earth system and ecological models often suffer from structural decouplings between spatial geometries and physical equations of state. Grid systems frequently treat spatial polygons merely as statistical geospatial containers or visual polygons rather than formal open thermodynamic control volumes. Consequently, cross-boundary advective transports and localized phase transitions risk non-physical mass leakage, violation of mass-energy closure, and implicit violations of the Second Law of Thermodynamics [5].

Sprint 042 of the *Web of Life* computational engine resolves this foundational challenge. Leveraging the Uber H3 Discrete Global Grid System (DGGs)—which provides an equal-area, low-distortion, spherical hexagonal tessellation of the globe [6]—we define an immutable, strongly-typed state tensor framework. In this paper, we document the mathematical formulation, architectural design, and invariant properties of the `H3CellThermodynamicState` interface and its supporting tensor classes.

2 Control Volume Specification over Hexagonal Prisms

Let $\mathcal{H}_r$ denote the set of discrete hexagonal cells on a sphere at H3 aperture resolution r . For each cell $c \in \mathcal{H}_r$, the control volume is defined as a prism with horizontal cross-sectional area A_c [m²], surface topography z_c [m], bounded inferiorly by the thermally active lithospheric/oceanic regolith layer depth d_{therm} , and superiorly by the top-of-atmosphere (TOA) boundary.

2.1 State Vector Definition

The complete thermodynamic state of cell c is characterized by the scalar state vector $\mathbf{x}_c \in \mathbb{R}^{22}$:

$$\mathbf{x}_c = [A_c, z_c, U_c, T_c, C_{v,c}, \alpha_c, \epsilon_c, \Phi_{\text{sw},c}^{\text{in}}, \Phi_{\text{sw},c}^{\text{out}}, \Phi_{\text{lw},c}^{\text{out}}, \Phi_{\text{sens},c}, \Phi_{\text{lat},c}, S_c, \sigma_c, M_d, M_w, M_l, M_i, M_v, M_C, M_N, M_P]^T \quad (1)$$

whose physical symbols, definitions, and SI units are detailed in Table 1.

Variable	Symbol	SI Unit	Physical Meaning
Area	A_c	m ²	Planar cross-sectional area of the H3 cell
Elevation	z_c	m	Surface elevation above mean sea level
Internal Energy	U_c	J	Total thermal sensible and latent energy content
Temperature	T_c	K	Absolute column surface temperature
Heat Capacity	$C_{v,c}$	J · K ⁻¹	Dynamic composite column isochoric heat capacity
Albedo	α_c	Dimensionless	Optical shortwave broadband surface reflectance
Emissivity	ϵ_c	Dimensionless	Longwave greybody surface thermal emissivity
Insolation Flux	$\Phi_{\text{sw},c}^{\text{in}}$	W · m ⁻²	Incoming top-of-atmosphere/surface solar flux
Reflected SW Flux	$\Phi_{\text{sw},c}^{\text{out}}$	W · m ⁻²	Shortwave flux reflected back to space ($\alpha_c \Phi_{\text{sw},c}^{\text{in}}$)
Outgoing LW Flux	$\Phi_{\text{lw},c}^{\text{out}}$	W · m ⁻²	Thermal blackbody radiation to space ($\epsilon_c \sigma T_c^4$)
Sensible Heat	$\Phi_{\text{sens},c}$	W · m ⁻²	Turbulent boundary layer sensible heat transfer
Latent Heat	$\Phi_{\text{lat},c}$	W · m ⁻²	Evaporative and sublimation turbulent heat transfer
Entropy	S_c	J · K ⁻¹	Absolute extensive thermal entropy of the column
Entropy Production	σ_c	W · K ⁻¹	Irreversible entropy production rate within the cell
Dry Air Stock	M_d	kg	Total column mass of dry atmospheric gases
Total Water Stock	M_w	kg	Total conserved water stock ($M_l + M_i + M_v$)
Liquid Water	M_l	kg	Surface and soil liquid water mass
Ice / Snow Water	M_i	kg	Solid phase water mass
Vapor Water	M_v	kg	Atmospheric column water vapor mass
Carbon Stock	M_C	kg	Total elemental carbon (biomass + soil + atmospheric)
Nitrogen Stock	M_N	kg	Total elemental nitrogen mass
Phosphorus Stock	M_P	kg	Total elemental phosphorus mass

Table 1: Scalar state properties defined in the `H3CellThermodynamicState` interface.

2.2 Fundamental Physical Invariants

Every instantiated state must strictly satisfy four invariant assertions prior to accepting state propagation:

1. Thermal Positivity:

$$T_c > 0 \text{ K} \quad \text{and} \quad U_c > 0 \text{ J} \quad (2)$$

2. Mass Stock Non-Negativity:

$$M_d \geq 0, \quad M_l \geq 0, \quad M_i \geq 0, \quad M_v \geq 0, \quad M_C \geq 0, \quad M_N \geq 0, \quad M_P \geq 0 \quad (3)$$

3. Water Mass Closure:

$$|M_w - (M_l + M_i + M_v)| \leq 10^{-7} \cdot M_w \quad (4)$$

4. Optical Range Restriction:

$$\alpha_c \in [0.0, 1.0], \quad \epsilon_c \in (0.0, 1.0] \quad (5)$$

3 First and Second Law Energy and Mass Evolution

3.1 The First Law of Thermodynamics: Energy Closure

The evolution of column internal energy U_c under continuous time t is governed by the boundary flux summation:

$$\frac{dU_c}{dt} = A_c [(1 - \alpha_c)\Phi_{\text{sw},c}^{\text{in}} - \epsilon_c \sigma T_c^4 - \Phi_{\text{sens},c} - \Phi_{\text{lat},c}] + \sum_{k \in \mathcal{N}(c)} J_{E,k \rightarrow c} \quad (6)$$

where $\sigma = 5.670374419 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$ is the Stefan-Boltzmann constant, $\mathcal{N}(c)$ is the set of topological neighbors ($|\mathcal{N}(c)| \leq 6$), and $J_{E,k \rightarrow c}$ represents the advective and diffusive energy flux across the shared boundary segment with neighbor k .

The dynamic composite heat capacity $C_{v,c}$ is calculated from the constituent masses:

$$C_{v,c} = M_d c_{v,d} + M_l c_w + M_i c_{\text{ice}} + M_v c_{v,\text{vap}} + M_{\text{lith}} c_{\text{soil}} \quad (7)$$

where $M_{\text{lith}} = \rho_{\text{lith}} A_c d_{\text{therm}}$ models the active thermal bedrock ($\rho_{\text{lith}} \approx 2600 \text{ kg/m}^3$, $d_{\text{therm}} = 2.0 \text{ m}$), and specific heat constants are set according to standard empirical determinations ($c_w = 4184 \text{ J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$, $c_{v,d} = 718 \text{ J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$).

The local temperature is rigorously derived from sensible internal energy:

$$T_c = \frac{U_{\text{sensible},c}}{C_{v,c}} = \frac{U_c - (M_l L_f + M_v (L_f + L_v))}{C_{v,c}} \quad (8)$$

where $L_f = 3.337 \times 10^5 \text{ J/kg}$ and $L_v = 2.501 \times 10^6 \text{ J/kg}$ designate the latent heats of fusion and vaporization, respectively.

3.2 The Second Law of Thermodynamics: Entropy Production

The total extensive entropy S_c satisfies the balance equation:

$$\frac{dS_c}{dt} = \frac{\dot{Q}_{\text{ext},c}}{T_c} + \sigma_c \quad (9)$$

where $\dot{Q}_{\text{ext},c}$ is the external net heat exchange, and σ_c is the local irreversible entropy production rate:

$$\sigma_c = \sigma_{\text{rad},c} + \sigma_{\text{phase},c} + \sigma_{\text{turb},c} + \sigma_{\text{diff},c} \quad (10)$$

Under the Second Law of Thermodynamics, every physical transition kernel within the *Web of Life* monad pipeline must enforce:

$$\sigma_c \geq 0 \quad \forall c \in \mathcal{H}_r \quad (11)$$

For solar thermal absorption, where shortwave photons emitted at $T_{\text{sun}} \approx 5778 \text{ K}$ degrade into terrestrial thermal kinetic energy at $T_c \approx 288 \text{ K}$, the local radiative entropy production rate is strictly positive:

$$\sigma_{\text{rad},c} = A_c (1 - \alpha_c) \Phi_{\text{sw},c}^{\text{in}} \left(\frac{1}{T_c} - \frac{1}{T_{\text{sun}}} \right) > 0 \quad (12)$$

because $T_{\text{sun}} \gg T_c > 0$.

4 Software Implementation in the Web of Life Architecture

The state tensor is implemented in TypeScript within `src/spatial/h3_state_tensor.ts`.

4.1 Object Model and Immutability

To avoid unexpected race conditions during parallel spatial computations, `H3CellThermodynamicRecord` is constructed as an immutable data object. Any physical modification creates a new instance via the `withUpdates()` copy-on-write method:

```
export class H3CellThermodynamicRecord implements H3CellThermodynamicState {
  constructor(
    public readonly h3Index: string,
    public readonly areaM2: number,
    public readonly elevationM: number,
    public readonly internalEnergyJ: number,
    public readonly temperatureK: number,
    public readonly heatCapacityJK: number,
    public readonly albedo: number,
    public readonly emissivity: number,
    public readonly shortwaveInWm2: number,
    public readonly shortwaveOutWm2: number,
    public readonly longwaveOutWm2: number,
    public readonly sensibleHeatFluxWm2: number,
    public readonly latentHeatFluxWm2: number,
    public readonly entropyJPerK: number,
    public readonly entropyProductionRateJKs: number,
    public readonly dryAirMassKg: number,
    public readonly totalWaterMassKg: number,
    public readonly liquidWaterMassKg: number,
    public readonly iceMassKg: number,
    public readonly vaporMassKg: number,
    public readonly carbonMassKg: number,
    public readonly nitrogenMassKg: number,
    public readonly phosphorusMassKg: number
  ) {
    this.assertInvariants();
  }

  public withUpdates(patch: Partial<H3CellThermodynamicState>): H3CellThermodynamicRecord {
    return new H3CellThermodynamicRecord(
      patch.h3Index ?? this.h3Index,
      patch.areaM2 ?? this.areaM2,
      ...
    );
  }
}
```

4.2 Tensor Aggregation Container

The collection of cells across a sub-grid or the planetary envelope is managed by `H3StateTensorContainer`. This container encapsulates a `Map<string, H3CellThermodynamicRecord>` and provides global reduction primitives that verify planetary-scale conservation:

$$M_{\text{total}} = \sum_{c \in \mathcal{H}_r} (M_{d,c} + M_{w,c} + M_{C,c} + M_{N,c} + M_{P,c}) \quad (13)$$

$$U_{\text{total}} = \sum_{c \in \mathcal{H}_r} U_c \quad (14)$$

These global operations provide an automated audit trail for continuous numerical verification during long integration runs.

5 Verification and Numerical Stability

Automated regression and invariant verification tests were integrated into the *Web of Life* test suite in `tests/sprint_042.test.ts`. The test suite asserts:

- **Invalid Construction Traps:** Instantaneous assertion errors upon encountering non-physical values ($T \leq 0$, negative elemental masses $M < 0$, or albedo outside $[0, 1]$).
- **Mass Conservation Under Partitioning:** Verifying that phase transitions (freezing, melting, vaporization) conserve exact mass within a relative tolerance $\epsilon_{\text{tol}} \leq 10^{-7}$.
- **Stefan-Boltzmann Radiative Accuracy:** Ensuring that $\Phi_{\text{lw}}^{\text{out}} = \epsilon\sigma T^4$ matches the expected theoretical value to within 10^{-6} W/m^2 .
- **Functional Monadic Purity:** Ensuring that transformations across time steps do not modify preceding state references.

6 Conclusion and Future Work

Sprint 042 establishes the canonical thermodynamic state representation for spatial modeling in the *Web of Life* simulation engine. By anchoring geodesic H3 hexagonal cells to formal thermodynamic control volumes, the platform establishes physical guarantees for planetary simulations.

Subsequent sprints (Sprint 043+) will build on this interface by implementing:

1. Hexagonal finite-volume advection solvers for atmospheric and oceanic shallow-water flows using shared hexagonal edge vectors;
2. Coupled Photosynthesis-Transpiration monads using Farquhar-Ball-Berry formulations directly bound to the carbon and latent heat fluxes of the state tensor;
3. Dynamic global entropy budgeting to quantify planetary maximum entropy production (MEP) states.

The full codebase and implementation details are openly available at the project repository: <https://github.com/pascalranoroarijaona/WebOfLife>.

References

- [1] A. Kleidon. *Thermodynamic Foundations of the Earth System*. Cambridge University Press, Cambridge, UK, 2016.
- [2] E. Schrödinger. *What is Life? The Physical Aspect of the Living Cell*. Cambridge University Press, Cambridge, UK, 1944.
- [3] E. N. Lorenz. Available potential energy and the maintenance of the general circulation. *Tellus*, 7(2):157–167, 1955.
- [4] G. W. Paltridge. Global dynamics and thermodynamics—a minimum entropy exchange principle. *Quarterly Journal of the Royal Meteorological Society*, 101(429):475–484, 1975.

- [5] H. Ozawa, A. Ohmura, R. D. Lorenz, and T. Pujol. The second law of thermodynamics and the global climate system: A review of the maximum entropy production principle. *Reviews of Geophysics*, 41(4):1018, 2003.
- [6] K. Sahr, D. White, and A. J. Kimerling. Geodesic discrete global grid systems. *Cartography and Geographic Information Science*, 30(2):121–134, 2003.